VPC Peering Connection

Project Introduction

1. Introduction

Amazon Virtual Private Cloud (Amazon VPC) allows resources to run inside logically isolated virtual networks. VPC Peering is a networking connection between two VPCs that enables resources in those VPCs to communicate using private IP addresses, provided that the required routes and security rules are configured.

This project presents a VPC Peering Connection architecture in an AWS Region. The diagram shows two VPCs, EC2 instances placed in subnets, route tables, an Internet Gateway, and a NAT Gateway. The purpose is to understand how network traffic is directed between virtual networks and how public and private subnet access can be managed.

2. Problem Statement

Applications deployed in separate VPCs may need to communicate privately without sending internal traffic over the public internet. The project explores how to establish private connectivity between VPCs and how routing and security configurations affect that connectivity.

3. Project Objectives

- Understand the role of Amazon VPCs, subnets, and IPv4 CIDR blocks.
- Illustrate communication between two VPCs using a VPC peering connection.
- Understand the purpose of route tables and routes for remote VPC CIDR blocks.
- Identify the roles of EC2 instances, an Internet Gateway, and a NAT Gateway in the shown design.
- Recognize that security groups and network routes must both permit traffic for connectivity to work.

4. Scope

The project focuses on the architecture and configuration concepts represented in the supplied diagram. A working implementation should verify the VPC CIDRs, subnet CIDRs, peering status, route targets, and security rules in the AWS Console before connectivity tests are reported.

5. Expected Learning Outcomes

On completion, the learner should be able to explain the purpose of VPC peering, describe how route tables direct traffic, distinguish internet access from private VPC-to-VPC connectivity, and identify the configuration checks required to troubleshoot failed communication.

6. Conclusion

VPC peering is one way to connect workloads in separate AWS VPCs through private IP addressing. A correct design requires non-overlapping VPC CIDR ranges, an active peering connection, routes on both sides, and appropriate security rules. The supplied diagram is a useful starting point for documenting and building this network.

Note: This introduction describes the architecture shown in the supplied image. It does not claim that the AWS resources have already been deployed or tested.